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Blockchain recall management in pharmaceutical industry

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Abstract

With the rapid development of the Internet and the Internet of Things, product recall, especially pharmaceutical recall, have attracted various stakeholders as a phase of product service. However, there are some the long-term problems on pharmaceutical recall, such as low effectiveness, seldom transparent, data can be tampered easily and so on. We have proposed a service system of pharmaceutical recall based on blockchain to solve these problems. The solution runs through the entire recall process that includes decision making, the OOS/OOT investigation, the action of working group and the Corrective and Preventative Action (CAPA). Through the management of recall, we are capable of shortening time, enhancing transparency and protecting the integrity of data in pharmaceutical recall.

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Keywords: blockchain network; CAPA;OOS/OOT, Product-Service System;

1. Introduction

Product recall is the process of retrieving defective and potentially unsafe goods from consumers while providing those consumers with compensation. Recall often occurs as a result of safety concerns over a manufacturing defect in a product that may harms its user. It will be the last stage of product service if product failure occurs.

With the improvement of consumption, product recall is attracting users for product safety in recent years.

Although product recall can be seen as the company's worst nightmare and short-term impact on shareholders' wealth is obvious [1], companies can mitigate the negative impact of product hazards by taking voluntary recall. In this way they also show their social responsibilities. [2]. Consumers will also change their attitudes from negative emotions after hearing product hazards to praise the companies' responsible actions.

This rule is enforced by all industries. But there are always opportunities who adopt a negative recall attitude. And some companies get poor evaluation because of the poor technology. So many problems would arise during the recall process.

Of all these problems, the time and transparency of recall will directly affect the company's reputation and the victimization of consumers especially in pharmaceutical industry.

Drug recall can be expensive for companies because they need to replace recalled products or pay for the caused loss [3]. Some companies choose to remain silence when drug harm occurs. The vaccine event of ChangSheng in 2018 brought risks to hundreds of thousands of children. Therefore, it is very important to find the drug hazard, make quick recall and show the transparent recall process.

The Internet, especially pharmaceutical websites, has become an important and preferred source of medicine and health information [4]. But these information is seldom transparent or easily tampering.

So we choose blockchain technology to solve these two problems. Blockchain is a distributed netbook technology with full network replication and network synchronization [5]. It has broad application prospect for many industries. Firstly, its data structure can record events in a verifiable and permanent way by an encryption algorithm. Its design is based on resistance to data modification. Once the transaction has been generated, its records cannot be recomposed. Secondly, the blockchain provides a useful trading mechanism based on smart contracts. Thirdly, the blockchain creates a new type of P2P communication network between members without the third

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party. Therefore, we try to use blockchain technology to create bridges in pharmaceutical recall.

The rest of the paper is organized as follows: Section 2 provides an architecture for pharmaceutical recall. The key steps of this architecture will be described carefully in Section 3. Finally, we will make a discussion in Section 4.

2. Architecture

We propose an architecture to shorten the time to start the recall after the primary signals of potential injuries [2] and enhance transparency during the process of product recall. There are three layers in this framework as figure 1. Generally, a product recall includes multiple steps of making decisions, taking actions to recall and Corrective and Preventative Action (CAPA). Participants cover multiple areas such as government, consumer and so on.

In player of the data source, there are four sources of the network of blockchain, the Centralized Database Management Systems (CDMS), the traditional Distributed Database Management Systems (DDMS), the Blockchain Database Management Systems (BDMS) or the data collected in real time. The last source consists of various smart assets, e.g. QR code, RFID tags [6] and readers, sensors GPS etc. The collected data will be delivered to the smart gateway. It plays the role of standardizing the raw data to enable the blockchain platform. The data and information provided from the perception layer will be transmitted to the component based on smart contract (CBSC) for generating the blocks.

The blockchain network layer contains Smart Contract, Consensus Protocol, Decentralized Application (DAPP) [7] and Cryptography etc. firstly, we use smart contracts to implement data conversion from the player of data source to blockchain. It provides the Component Based on Smart Contract (CBSC), which has the function of integrating and transmitting information in one party (node). The data is uploaded to the blockchain network according to a predefined smart contract to implement distributed storage and ensure that the data is not tampered. CBSC also acts on the next layer. Secondly, the consensus protocol play a core role in the system. It is the algorithm to reach the agreement among nodes in the blockchain network. Thirdly, DApp is deployed on a distributed network compared to previous apps running on a centralized server. All data is transparent and tamper-proof. Finally, all uploaded data are encrypted, and different keys are used for encryption and decryption.

The application layer is composed of a series of systems, services and software products provided by the companies and the management component is provided by us. They include the Out of Specification (OOS) [8] management system, Out of Trend (OOT) [9] management system, Supply Chain Management Systems (SCMS) [10], Logistics Management Systems (LMS) [11], software for Enterprise Resource Planning (ERP) and many other proprietary software and systems. The application layer works closely with the blockchain network through the CBSC. Certainly, the data generated by the application layer will also be uploaded as a data source to the blockchain network under the consensus protocol.

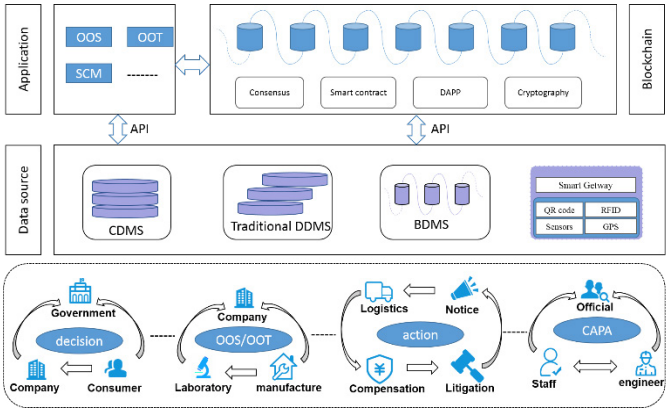


Fig. 1. The architecture of the pharmaceutical recall management system based on blockchain

Under this architecture, the entire process of the recall will be monitored. And the data is more transparent to all stakeholders through the blockchain network.

3. Key steps

3.1. Synergistic decision

In the process of decision making, we need to get the results of the type and the level of product recall.

There are two types of product recall, the proactive recall [12] and the passive recall. The proactive recall is that drug manufacturers recall drugs with potential safety hazards before the official compulsory through the collection and analysis of information, investigation and evaluation. The passive recall is that the regulation parts of government commands the drug manufacturers to recall the drugs having potential safety hazards via investigation and evaluation. Before the related parts of government take measures, the drug manufacturers don't take any actions for drug failures. We support automatic data analysis services to help government and drug manufacturers make decision quickly.

There are three levels of product recall. The first level is to recall products that may cause serious health hazards. The second level is to recall product that may cause temporary or reversible health hazards. The third level is to recall products generally do not cause health hazards, but they needs to be recalled for other reasons. Generally the recall measures will be different under the different levels. We support the service to analyze the collected data to evaluate the level of recall.

Many quality issues can lead to recall decisions, such as bias, Investigating Out of Specification (OOS), Investigating Out of Trend (OOT), complaints, adverse drug reactions and so on. The step of synergistic decision making starts by reading data in real time from the network of blockchain as Figure 2. Then our system will make quality issues detection based on the data. The result of quality issues detection will trigger the smart contract to generate a recall level or nothing. Once the recall level is made, the corresponding recall plan will also be generated.

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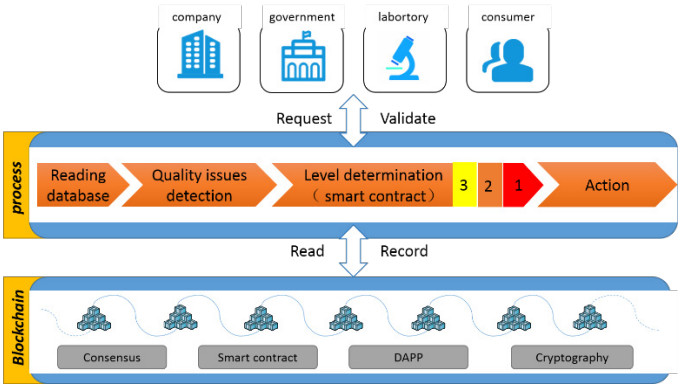


Fig. 2. Synergistic decision making

Some key processes of Synergistic decision making are as follows.

3.1.1 Quality issues detection

In this process, we have design an analyzed model to get the deviation, the complaint rate and adverse events as the Figure 3.

Then the result will be identified by the smart contract. The smart contract will command the laboratory of related companies or governments to finish the OOS/OOT investigation when the result meet the recall conditions. All records in this process need be uploaded to the network of blockchain.

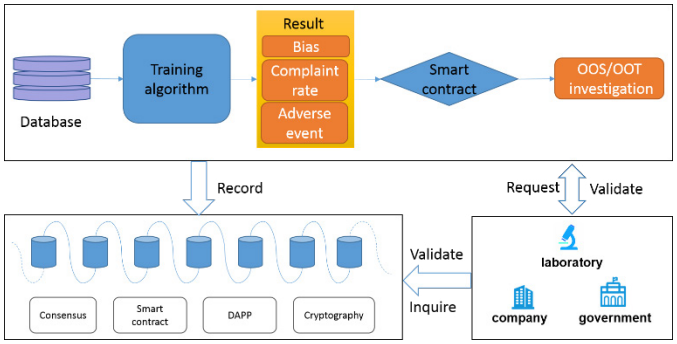


Fig. 3. Quality issues detection

Although the drug testing standards vary from region to country, the dimensions of the test are consistent. Test items include temperature, acidity, drug composition, etc.

Suppose a drug test data has **m** dimensions, and a sample has **n** groups of data.

$$x_i = [x_{i1}, x_{i2}, x_{i3}, \cdots x_{im},](i = 1,2,3\cdots, n) \tag{1}$$

The standard deviation is calculated as follows.

$$S_1 = f(x_i) \tag{2}$$

S_1—the standard deviation (%)

The types of deviations in actual medical testing are not limited to standard deviations. We assume that the set of deviations for a drug is S.

$$S = [S_1, S_2, S_3, \cdots] \tag{3}$$

The complaint rate is calculated as follows.

$$cr = \frac{m}{n} \times \% \tag{4}$$

cr—the complaint rate (%)

m—the number of complaints

n—total number of events reflected by consumers

Product recalls and OOS/OOT investigation will be implemented once complaint rate from real-time monitoring exceed the standard requirements.

The adverse events will directly lead to recall and OOS/OOT investigation.

3.2. The OOS/OOT investigation

The OOS/OOT investigation is shown as Figure 4. Once the decision of OOS/OOT investigation is made, we need do the laboratory analysis to get the OOS/OOT result. The smart contract will decide whether to proceed to the next phase of investigation based on this result. Under the control of smart contracts, the OOS survey will go through multiple phases.

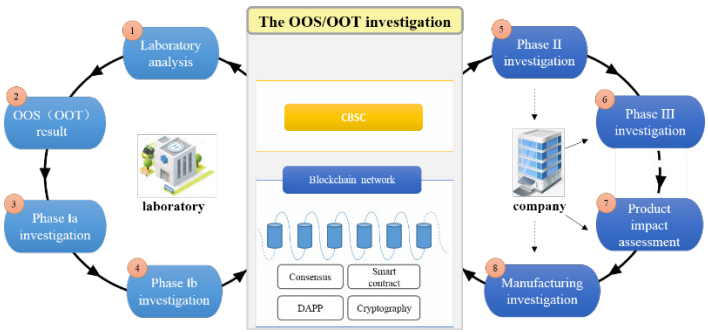


Fig. 4. The OOS/OOT investigation

3.2.1 Laboratory analysis

In the course of laboratory analysis, these measures have to be done as follows.

- Batch release testing and testing of starting materials
- In-Process Control testing.
- Stability studies on marketed batches of finished products, active pharmaceutical ingredients.
- Previous released batch used as reference sample in an OOS investigation showing OOS or suspect results.
- Batches for clinical trials.

Then all solutions and reagents must be retained until all data has been verified by the second person as being within the defined acceptance criteria. However if the sample test criteria is usually the first level of testing and a sample has to be tested to the next level this should be investigated as it is not following the normal trend.

The results of an OOS investigation are as follows.

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- Test results don't comply with the predetermined acceptance criteria.
- Test results fall outside of established acceptance criteria which have been established in official compendia or company documentations.

The results of an OOT investigation are as follows.

- Generally a stable results don't follow the expected trend
- The trends of starting materials and in-process samples may also yield out of trend data. The results are not necessarily OOS, but they are not typical data points.

The atypical, aberrant and anomalous results are as follows.

- Results that are still within specification but are unexpected, questionable, irregular, deviant or abnormal.

Then the results of an OOS/ OOT investigation will be identified by CBSC. The CBSC will determine whether the next phase of the investigation is needed. At the same time, the data generated during this phase will be uploaded to the blockchain network under the consensus protocol.

3.2.2 The phase I of investigation

If the products is Out of Specification (OOS) or Out of Trend (OOT) in the phase of laboratory analysis, the phase Ia of investigation will be done. In the phase Ia, we need to find the obvious errors as follows.

- Calculation error
- Power outage
- Equipment failure
- Testing errors
- Incorrect Instrument Parameters

If no error was noted and none of the above conditions were met, phase Ib of investigation must be taken place. It's an initial investigation conducted by the analyst and supervisor with the Laboratory Investigation Checklist. And it should only be restricted to data, equipment and analysis review. The checklist may not be all-inclusive, but it should be a good guideline to cover the pertinent areas in any laboratory investigation.

Then the results of this phase will be identified by the component based on the smart contract. This component will determine if the next phase of the investigation is needed. At the same time, the data generated during this phase will be uploaded to the blockchain network under the consensus protocol.

3.2.3 The phase II of investigation

Phase II will be conducted when the phase I investigations do not reveal an assignable laboratory error. Phase II investigations are driven by written and approved instructions for hypothesis. Prior to further testing, a manufacturing investigation should be started to determine whether there was a possible manufacturing cause. When considering performing additional testing, it is important to perform a predefined retesting plan by an analyst other than the one who performed the original test. The steps are as follows.

- Hypothesis/Investigative Testing: to confirm or discount possible root causes.
- Re-Test: Performing the test over again by using material from the original sample composite if it is still available. Otherwise a new sample will be used.
- Re-sample: to use a new sample from the original container if there is insufficient material remaining in the original sample composite or a problem with the integrity of the original sample.
- Most probable consequence: to get the result scientifically and reasonably.

Then the results of this phase will be identified by the component based on smart contract. This component will determine if the next phase of the investigation is needed. At the same time, the data generated during this phase will be uploaded to the blockchain network under the consensus protocol.

3.2.4 The phase III of investigation

To determine whether other batches of product are affected and make identification and implementation of corrective and preventative action (CAPA), an investigation (phase III) still needs to be performed though the phase II has been finished and the batchs of product have been recalled.

The investigation of this phase should be started by reviewing the completed manufacturing surveys and combining the incorporate suspicious laboratory surveys or possible causes into results after the validation. To conclude the investigation, all of the results must be evaluated. The investigation report should contain a summary of the investigations performed and a detailed conclusion.

According to the previous survey, the results of the third phase are as follows.

- If no laboratory or calculation errors are found in phase I and phase II, there is no scientific basis to invalidate the initial OOS results.
- All test results should be reported (in all QC documents and any analysis certificates) and all data must be considered in the bulk release decision.
- If the investigation determines that the initial sampling method is inherently inadequate, then a new accurate sampling method must be developed and documented.
- The initial OOS result does not necessarily mean that the subject batch failed and must be rejected. The OOS results should be investigated.

The OOS/OOT investigation play core role in a product recall. It will directly affect the decision of the recall working group and the judgment of the CAPA improvement team described in later sections.

3.3. Coordination action

The action of the recall is divided into the preparation phase and the implementation phase of the program.

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3.3.1 The preparation phase

On the technical side, we need to make the following preparations.

- List the batch number, quantity, sales quantity, inventory quantity of the products involved in the recall.
- Freeze materials and products related to recalled products;
- List the list of customers who need to notify the recall. The list should include the specific contact information and the specific address of the product;
- Preparation of alternative supply options;
- Initially determine the measures to return, collect, coordinate and ultimately destroy.

On the communication side, we need to make the following preparations.

- Make different forms of reports, notices for different communication objects (e.g. drug regulatory authorities, customers, company employees, partners, public media (if necessary));
- Make solutions for foreseeable external problems;
- Determine different communication leaders and contact information for different communication objects;
- Select communication media such as conference, telephone, fax machine, E-mail, public media (television announcement, broadcast)

On the economic side, we need to make the plan of customer compensation [13], corresponding funds and other financial preparations.

Meanwhile，we need do corresponding legal preparation.

In our framework, the preparation will be carried out as figure 5.

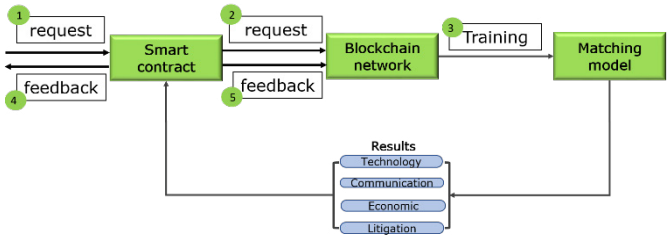


Fig. 5. The preparation phase

Firstly, the request from the working group will be sent to the component based on the smart contract. Secondly, this component will process the request and send a request to the blockchain network to get the corresponding data. Thirdly, the matching model will train the data to generate initial plans. The plan contains four sides mentioned above. Finally, the component based on the smart contract will send these plans to the working group for selecting, and feedback all records to blockchain network.

3.3.2 Implementation phase

In this phase, the working group will implementation plan made in last phase as Figure 6.

On the technical side, their work includes notifying users, acceptance, materials balance and alternative supply. On the communication side, they need to communicate with governments, customers, media and staff. On the economic side, they need to negotiate compensation and then complete compensation. Finally, they need to cope with litigation.

Then all records of this phase will be identified and uploaded to the blockchain network under the consensus protocol.

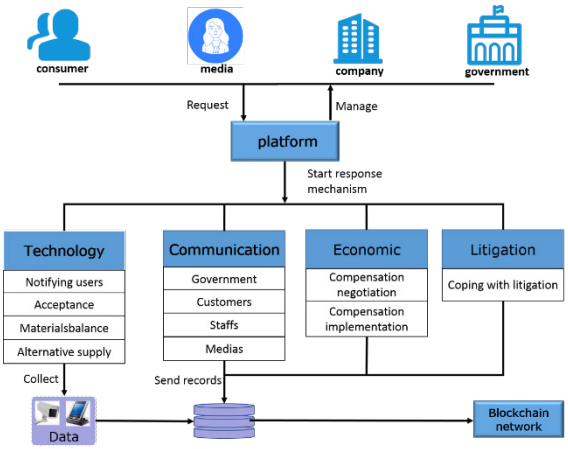


Fig. 6. The implementation phase.

3.4. CAPA based on CBSC

The traditional CAPA [14] is mainly based on QA engineers and CAPA administrators. Now we propose a new solution through the component based on smart contract (CBSC).

There are three phase about CAPA in our system as figure 7. The participants of proposed phase include QA engineers and blockchain network of our system. The CBSC monitors the data uploaded in real time. Once the error happens, the notice will be sent to QA engineers. Then QA engineers will submit application to CBSC for review.

In improvement phase, a preliminary investigation will be taken first and remedial measures will also be taken simultaneously. And then an in-depth investigation will be launched. This investigation contains the OOS/OOT investigations mentioned in 3.2.

The data of this investigation will be upload to CBSC. This CBSC will send a request to the blockchain network to get the corresponding data. Then the matching model will train the data to generate initial CAPA plans.

The improvement team will make preliminary verification for initial CAPA plans and select an acceptance plan for review. If the result from QA engineers is fail, the in-depth investigation will be continued. Otherwise, the CAPA plan will be implemented. Then the QA engineers will take the implementation of CAPA. If the result is no, the CAPA plan will be implemented again. If the result is yes, the related data will be upload to CBSC for reviewing.

In the acceptance phase, the CBSC will make a decision whether the CAPA plan will be accepted. If it is permitted, the system will restart CAPA.

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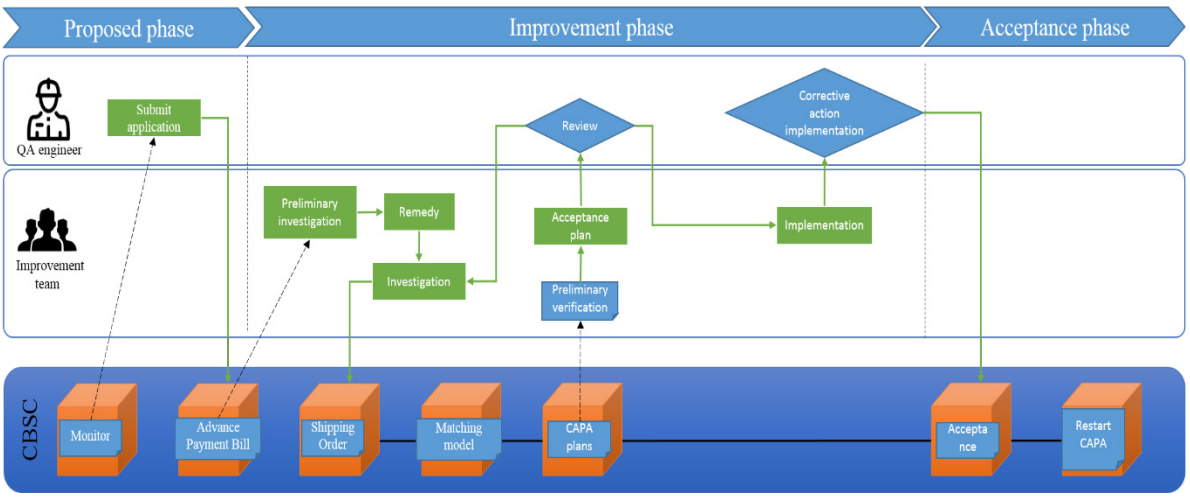


Fig. 7. The CAPA based on CBSC.

4. Discussion

This paper has outlined a framework within which various stakeholders are encouraged to participate in open management in the pharmaceutical recall. This framework consists of a smart gateway for getting multi-object data, a component based on smart contract (CBSC) for the party to gather, process and regulate the pharmaceutical recall, a blockchain network and various applications. Through this management of recall, we can shorten the time, enhance transparency and protect the integrity of data in product recall.

There are already many websites for medical recall on the market. Most of them do not have full open management and resistance of tampering. In the future, open management of recall will be the general trend, but it takes certain time.

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